NXTVOLT Procurement

Database guide: the tables the module uses, how they are linked, and how data moves between them

Database	nxtvolt (PostgreSQL 16, in the Docker service db)
Odoo module	procurement_custom, version 20.0.1.2.0
Generated	14 October 2026 from the live database
Data at that moment	3 client POs, 73 order lines, 28 quotes, 9 purchases, 6 delivery groups, 6 payment milestones

Contents

1. How Odoo turns code into tables
2. Map of the tables (diagram)
3. The tables one by one
4. How the tables communicate: the life of one item
5. Automatic calculations and the rules that protect the data
6. What happens on delete
7. Useful SQL queries

Appendix: every column the module adds

1. How Odoo turns code into tables

You never create tables by hand. Each Python class in `addons/procurement_custom/models/` is an Odoo **model**, and when the module is installed or upgraded Odoo creates or alters the matching PostgreSQL table.

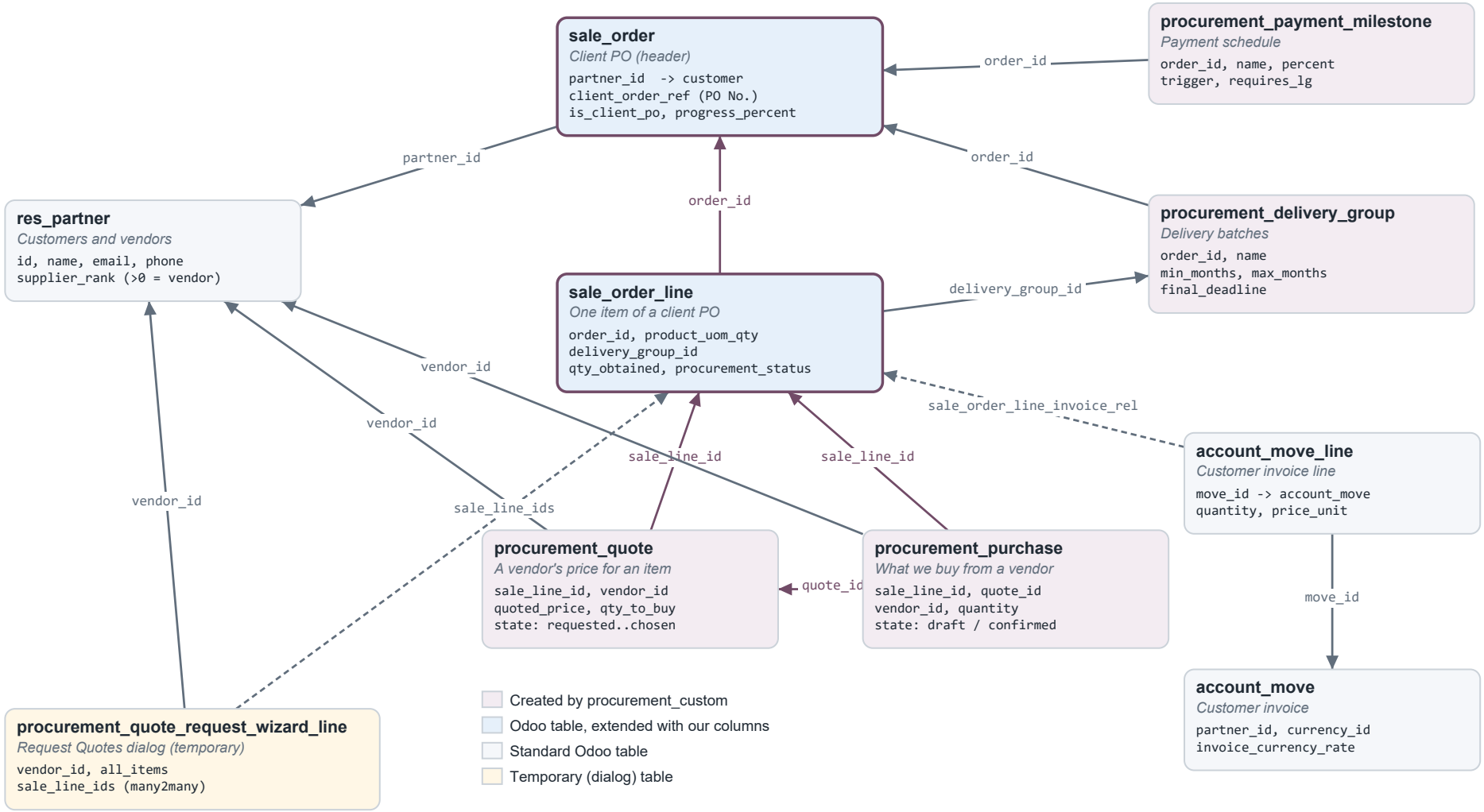
In the Python code	In the database
Model `procurement.quote`	Table `procurement_quote` (dots become underscores)
`_inherit = "sale.order.line"` (no new _name)	No new table: our fields become **extra columns** of Odoo's existing `sale_order_line`
Simple field (Char, Float, Date, Selection...)	One column
`Many2one` (e.g. `vendor_id`)	One integer column holding the other row's id, with a real FOREIGN KEY
`One2many` (e.g. `quote_ids` on an order line)	**No column.** It is the reverse view of a Many2one on the other table
`Many2many` (e.g. the wizard's `sale_line_ids`)	A separate link table with two id columns, one row per pair
`related=..., store=True`	A copied column, kept in sync by Odoo (e.g. `procurement_quote.order_id`)
`compute=..., store=True`	A column Odoo recalculates itself when its inputs change
`compute=...` without store	No column: calculated each time it is displayed
`TransientModel` (wizard)	A table whose rows Odoo deletes automatically after a while

Every table also gets `id` (primary key), `create_uid`/`create_date` (who and when created) and `write_uid`/`write_date` (who and when last changed). They are left out of the rest of this guide.

Of the roughly 440 tables in the database, the module **created 7** (all starting with `procurement_`) and **added columns to 2** of Odoo's: `sale_order` and `sale_order_line`. The rest belong to Odoo's apps (Sales, Invoicing, Contacts, Products, the core system) and are used as they are.

2. Map of the tables

Each box is a table; each arrow is a link stored in the table it starts from, pointing to the row it refers to. The label is the column holding the link. Dashed arrows go through a separate link table. Purple arrows are the main procurement chain.



Not drawn, to keep the map readable: `currency_id` on quotes and purchases (-> `res_currency`), `product_id` on order lines (-> `product_product`), `order_id` copied onto quotes and purchases (-> `sale_order`), and the Request Quotes dialog's header table `procurement_quote_request_wizard`.

3. The tables one by one

Only the important columns are listed here. The appendix lists every column the module adds. **FK** marks a foreign key (a link to another table).

sale_order: the client PO

Odoo table (Sales app), extended by the module

One row per order. A client PO is an Odoo sales order with `is_client_po = true`; the module adds everything printed on the client's PDF and the procurement progress.

Column	Type	Meaning
`name`	text	Our reference, e.g. S00046
`partner_id`	FK res_partner	The customer (client)
`client_order_ref`	text	The client's PO number, e.g. 65282
`pricelist_id`	FK product_pricelist	Decides the PO currency (USD for the demo PO)
`amount_untaxed / amount_tax / amount_total`	money	Totals Odoo calculates from the lines
`is_client_po`	yes/no	Marks a sales order as a client PO (ours)
`client_po_date, client_pr_number, ...`	various	Header fields as printed on the PO (ours)
`*_printed`	various	Values copied exactly as printed, kept to check against Odoo's own
`totals_match`	yes/no, computed	Printed totals equal Odoo's (within 0.01)
`item_count / items_chosen_count / items_obtained_count`	number, computed	How many items, how many have a vendor, how many are fully obtained
`progress_percent`	number, computed	Share of items fully obtained
`advance_payment_received_date`	date	Starts the delivery deadlines

sale_order_line: one item of a PO

Odoo table (Sales app), extended by the module

One row per item (and per section/note heading, which have `display_type` set). This is the **hub** of the module: quotes and purchases all point at a line.

Column	Type	Meaning
`order_id`	FK sale_order	The PO this item belongs to
`product_id`	FK product_product	The product
`product_uom_qty`	number	Quantity the client ordered
`price_unit, price_subtotal`	money	Client's unit price and line total
`display_type`	text	Empty for a real item; 'line_section' / 'line_note' otherwise
`client_line_no`	number	S. No. on the client's PO (unique within a PO)
`client_item_code, description_printed, uom_printed, total_printed`	text / money	The line exactly as printed
`delivery_group_id`	FK procurement_delivery_group	Delivery batch of the item

Column	Type	Meaning
chosen_vendor_id	FK res_partner, computed	First chosen vendor (for reports)
qty_obtained / qty_remaining	number, computed	Bought so far (confirmed purchases) / still to buy
avg_purchase_price, margin_amount	money, computed	Average cost and margin, in the PO currency
procurement_status	choice, computed	not_quoted, quote_requested, quoted, vendor_chosen, partially_obtained, obtained

procurement_quote: a vendor's price for an item

Created by the module

One row per (item, vendor) pair. The same item can be asked of several vendors, and a vendor can be asked for several items. The pair (sale_line_id, vendor_id) is unique.

Column	Type	Meaning
sale_line_id	FK sale_order_line	The item quoted
vendor_id	FK res_partner	The vendor
order_id, client_line_no	copied	Copied from the item, for grouping and sorting
state	choice	requested -> quoted -> chosen, or rejected
quoted_price, currency_id	money, FK res_currency	The vendor's unit price and its currency
price_order_currency	money, computed	Price converted to the PO currency, to compare vendors
is_lowest	yes/no, computed	Cheapest priced quote of the item
qty_to_buy	number	How many units to buy from this vendor
amount_to_buy	money, computed	qty_to_buy x quoted_price
qty_needed	number, not stored	Ordered quantity not yet assigned to a chosen vendor

procurement_purchase: what we buy from a vendor

Created by the module

One row per purchase. An item can have several (split between vendors, or bought in parts). Choosing a quote creates one in draft; only **confirmed** rows count as obtained.

Column	Type	Meaning
sale_line_id	FK sale_order_line	The item bought
quote_id	FK procurement_quote	The quote whose choice created it (empty if typed by hand)
vendor_id	FK res_partner	The vendor
state	choice	draft (shown as To Confirm) or confirmed
quantity, purchase_price, currency_id	number, money, FK	What was bought, at what price
amount	money, computed	quantity x purchase_price
date	date	Purchase date; also the date of the exchange rate used
qty_received	number	Received from the vendor (not used by any calculation yet)

procurement_delivery_group: delivery batches

Created by the module

Groups a PO's items that are delivered together, with their deadlines.

Column	Type	Meaning
`order_id`	FK sale_order	The PO
`name, description, boq_reference`	text	Identification of the batch
`min_months, max_months, start_event`	number, choice	Delivery window
`immediate_batch, immediate_batch_weeks`	yes/no, number	Optional first batch
`immediate_batch_deadline, final_deadline`	date, computed	From the PO's advance_payment_received_date

Items point at their group through `sale_order_line.delivery_group_id`; the group's `line_ids` is the reverse view of that column.

procurement_payment_milestone: payment schedule

Created by the module

The PO's payment terms, split into milestones. Their percentages may not exceed 100 in total.

Column	Type	Meaning
`order_id`	FK sale_order	The PO
`name, percent`	text, number	e.g. 'Advance', 25
`trigger, collection_days`	choice, number	When it becomes due
`requires_lg, lg_type`	yes/no, text	Letter of guarantee needed for this payment

res_partner: customers and vendors

Standard Odoo table (Contacts)

Every person and company. The same table holds clients and vendors; a vendor has `supplier_rank > 0`.

Column	Type	Meaning
`name, email, phone, street, city, country_id`	text, FK	Contact details
`is_company`	yes/no	Company or person
`supplier_rank / customer_rank`	number	Above 0 = used as vendor / customer

Referenced by sale_order.partner_id (client), the buyer/requester on the PO, and vendor_id on quotes, purchases and the Request Quotes dialog.

account_move and account_move_line: customer invoices

Standard Odoo tables (Invoicing)

An invoice is a row of `account_move`; its product rows are in `account_move_line`. Invoice lines are linked to order lines through the link table `sale_order_line_invoice_rel`, which is how Odoo knows each item's invoiced quantity.

Column	Type	Meaning
`account_move.partner_id`	FK res_partner	The customer invoiced
`account_move.state`	choice	draft, posted, cancel
`account_move.currency_id`	FK res_currency	Invoice currency
`account_move.invoice_currency_rate`	number	The rate box next to the currency; the module converts line prices with it
`account_move_line.move_id`	FK account_move	The invoice
`account_move_line.quantity, price_unit`	number, money	Invoiced quantity and price
`sale_order_line_invoice_rel`	link table	Pairs (order_line_id, invoice_line_id); drives sale_order_line.qty_invoiced

res_currency and res_currency_rate: currencies

Standard Odoo tables

Currencies and their dated exchange rates. Used to compare quotes in different currencies, to convert purchase costs to the PO currency, and to convert invoice prices.

Column	Type	Meaning
`res_currency.name`	text	USD, EGP...
`res_currency_rate.currency_id, name, rate`	FK, date, number	Units of the currency per one unit of the company currency, from that date on

Request Quotes dialog tables

Created by the module, temporary

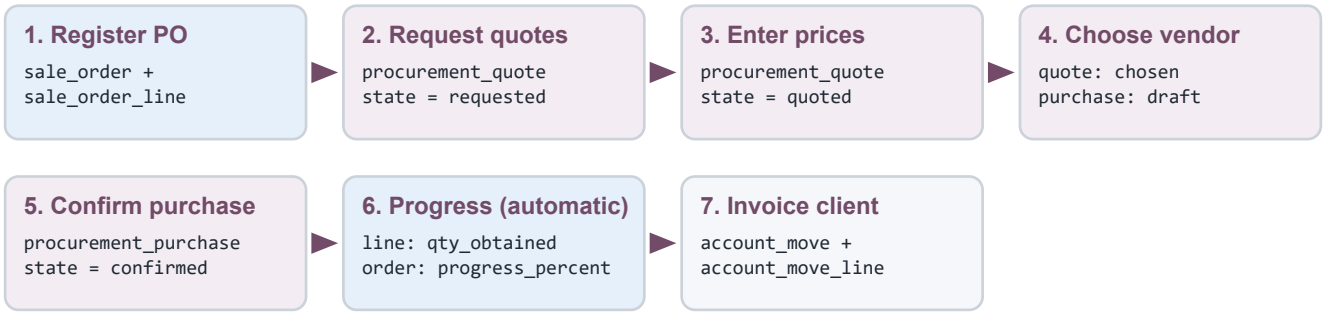
Hold what is typed in the Request Quotes dialog until its button creates the quotes; Odoo then clears them automatically.

Column	Type	Meaning
`procurement_quote_request_wizard`	table	The dialog: `order_id` (FK sale_order)
`procurement_quote_request_wizard_line`	table	One vendor row: `wizard_id`, `vendor_id`, `all_items`
`..._wizard_line_sale_order_line_rel`	link table	The items picked for that vendor

An empty table `procurement_quote_choose_wizard` is left from a dialog that was added and later removed. Odoo does not drop old tables; it is unused and harmless.

4. How the tables communicate: the life of one item

The tables do not talk to each other directly. Buttons and saved edits in the Odoo screens run the module's Python code, which writes rows; Odoo then recalculates the computed columns that depend on what changed. Here is what each step writes, using an item ordered 10 times.



Step (screen)	What is written	What Odoo then recalculates
1. Register the PO Client POs > New	1 row in sale_order (is_client_po = true, partner_id = client) 1 row per item in sale_order_line (order_id)	Totals, totals_match, line_total_match, item_count
2. Request quotes PO > Request Quotes	Dialog rows in the wizard tables, then 1 row per (item, vendor) in procurement_quote, state = requested	Item status: quote_requested
3. Enter a price Quote comparison	procurement_quote.quoted_price; the module sets state = quoted and quote_date	price_order_currency, is_lowest of all quotes of the item; item status: quoted
4. Choose a vendor Qty to Buy, then Choose	procurement_quote: state = chosen, qty_to_buy = 6 New procurement_purchase: quote_id, sale_line_id, vendor_id, quantity 6, price, state = draft If 10 of 10 are assigned: the other open quotes get state = rejected	qty_needed on the item's quotes (10 -> 4); chosen_vendor_id; item status: vendor_chosen; items_chosen_count on the PO
4b. Change a chosen quantity Edit Qty to Buy	procurement_quote.qty_to_buy; the linked draft purchase's quantity follows; rejected quotes reopen if quantity is free again	amount_to_buy, purchase amount, qty_needed
5. Confirm the purchase Purchases > Confirm	procurement_purchase.state = confirmed	On the item: qty_obtained, qty_remaining, avg_purchase_price, margin_amount, procurement_status (partially_obtained / obtained) On the PO: items_obtained_count, progress_percent
Undo a choice Quote comparison > Undo	Deletes the quote's draft purchase; quote back to quoted; rejected quotes reopen	Same as step 4, in reverse. Refused if the purchase is already confirmed
6. Invoice the client PO > Create Invoice	1 row in account_move, 1 per item in account_move_line, pairs in sale_order_line_invoice_rel	sale_order_line.qty_invoiced. Changing the invoice currency or its rate re-converts the line prices

5. Automatic calculations and the rules that protect the data

Computed columns form a chain. When a row at the bottom changes, Odoo walks up the chain and recalculates every stored column that depends on it, in the same transaction.

```
procurement_purchase (state, quantity, purchase_price, currency_id, date)
procurement_quote     (state, qty_to_buy, quoted_price)
        |
        v
sale_order_line   qty_obtained, qty_remaining, avg_purchase_price,
                  margin_amount, procurement_status, chosen_vendor_id
        |
        v
sale_order        items_chosen_count, items_obtained_count, progress_percent
```

Rules enforced by the module

Rule	Where	Enforced by
One quote per (item, vendor)	procurement_quote	UNIQUE constraint in PostgreSQL
Chosen quantities of an item <= ordered quantity	procurement_quote	Python constraint
qty_to_buy >= 0; purchase quantity > 0; purchase price >= 0	quote / purchase	CHECK constraints
A confirmed purchase locks its quote's quantity and blocks Undo	quote	Python (write / Undo)
S. No. unique within a PO	sale_order_line	Partial UNIQUE index (only where S. No. > 0)
Delivery group must belong to the item's own PO	sale_order_line	Python constraint
Payment milestones total at most 100 %	procurement_payment_milestone	Python constraint

6. What happens on delete

Each foreign key says what PostgreSQL does to the rows pointing at a row that is deleted. These are the actual settings in the database:

When this is deleted	These rows	Effect
A client PO (sale_order)	its delivery groups, payment milestones, order lines	Deleted too (CASCADE)
An order line	its quotes and purchases	Deleted too (CASCADE)
A quote	the purchases created from it	Kept; their quote_id is emptied (SET NULL)
A delivery group	the order lines in it	Kept; their delivery_group_id is emptied (SET NULL)
A vendor (res_partner) or currency	quotes and purchases using it	Refused while any exists (RESTRICT)

Odoo normally prevents deleting a confirmed sales order from the screens; it must be cancelled first.

7. Useful SQL queries

For reading only, in pgAdmin (http://localhost:5050) Query Tool or `docker compose exec db psql -U odoo -d nxtvolt`. Change data through Odoo, not SQL: SQL changes skip the recalculations and rules above.

Client POs with their customer

```
SELECT o.name AS our_ref, o.client_order_ref AS po_no, c.name AS customer,
       o.amount_total, o.progress_percent
FROM sale_order o
JOIN res_partner c ON c.id = o.partner_id
WHERE o.is_client_po;
```

Every quote of one PO, item by item

```
SELECT l.client_line_no, l.client_item_code, v.name AS vendor,
       q.quoted_price, q.qty_to_buy, q.state
FROM procurement_quote q
JOIN sale_order_line l ON l.id = q.sale_line_id
JOIN res_partner v     ON v.id = q.vendor_id
JOIN sale_order o      ON o.id = l.order_id
WHERE o.name = 'S00046'
ORDER BY l.client_line_no, q.price_order_currency;
```

Purchases waiting for confirmation

```
SELECT o.name, l.client_line_no, v.name AS vendor, p.quantity, p.amount
FROM procurement_purchase p
JOIN sale_order_line l ON l.id = p.sale_line_id
JOIN sale_order o      ON o.id = l.order_id
JOIN res_partner v     ON v.id = p.vendor_id
WHERE p.state = 'draft';
```

Ordered, obtained and invoiced per item

```
SELECT o.name, l.client_line_no, l.product_uom_qty AS ordered,
       l.qty_obtained, l.qty_invoiced, l.procurement_status
FROM sale_order_line l
JOIN sale_order o ON o.id = l.order_id
WHERE o.is_client_po AND l.display_type IS NULL
ORDER BY o.name, l.client_line_no;
```

Appendix: every column the module adds

Taken from Odoo's field registry. **Stored = no** means the value is calculated when shown and has no column; one2many fields have no column either (they read the other table's FK).

sale_order (columns added)

Field (column)	Type	Links to	Stored	Label in Odoo
is_client_po	boolean		yes	Client PO
client_name_printed	char		yes	Client Name (as printed)
client_address_printed	text		yes	Client Address (as printed)
client_po_box	char		yes	P.O. Box
client_phone_printed	char		yes	Phone (as printed)
client_fax_printed	char		yes	Fax (as printed)
client_po_date	date		yes	PO Creation Date
client_po_approval_date	date		yes	PO Approval Date
client_pr_number	char		yes	PR No.
client_pr_approval_date	date		yes	PR Approval Date
client_buyer_id	many2one	res_partner	yes	Buyer
client_buyer_name_printed	char		yes	Buyer (as printed)
client_requester_id	many2one	res_partner	yes	Requester
client_requester_name_printed	char		yes	Requester (as printed)
client_requester_dept	char		yes	Requester Dept.
client_po_type	char		yes	PO Type
rfp_reference	char		yes	RFP Reference
penalty_text_printed	char		yes	Penalty (as printed)
penalty_rate_weekly	float		yes	Penalty per Week (%)
penalty_cap	float		yes	Penalty Cap (%)
penalty_base	selection		yes	Penalty Base
supplier_name_printed	char		yes	Supplier Name (as printed)
supplier_address_printed	text		yes	Supplier Address (as printed)
payment_term_text_printed	char		yes	Payment Term (as printed)
delivery_term_printed	char		yes	Delivery Term (as printed)
delivery_time_printed	char		yes	Delivery Time (as printed)
delivery_location_printed	char		yes	Delivery Location (as printed)
payment_terms_text_printed	text		yes	Payment Terms (as printed)
payment_milestone_ids	one2many	procurement_payment_milestone	no	Payment Milestones
payment_currency_id	many2one	res_currency	yes	Payment Currency
exchange_rate_rule	char		yes	Exchange Rate Rule
payment_counter_start	char		yes	Payment Counter Start
lg_release_condition	char		yes	LG Release Condition
partial_invoicing_allowed	boolean		yes	Partial Invoicing Allowed
pac_fac_applied	boolean		yes	PAC/FAC Applied

Field (column)	Type	Links to	Stored	Label in Odoo
advance_payment_received_date	date		yes	Advance Payment Received Date
delivery_terms_text_printed	text		yes	Delivery Terms (as printed)
delivery_group_ids	one2many	procurement_delivery_group	no	Delivery Groups
delivery_destination_printed	char		yes	Delivery Destination (as printed)
warranty_text_printed	text		yes	Warranty (as printed)
warranty_months	integer		yes	Warranty Months
warranty_start	selection		yes	Warranty Starts From
warranty_scope	char		yes	Warranty Scope
warranty_reference	char		yes	Warranty Reference
notes_to_supplier_printed	text		yes	Notes to Supplier (as printed)
annex_reference	char		yes	Annex Reference
approved_by_name	char		yes	Approved By Name
approved_by_department	char		yes	Approved By Department
client_stamp_present	boolean		yes	Client Stamp Present
other_stamps_note	text		yes	Other Stamps
handwritten_marks	char		yes	Handwritten Marks
invoice_submission_text_printed	text		yes	Invoice Submission (as printed)
invoice_submission_days	char		yes	Invoice Submission Days
invoice_submission_location	char		yes	Invoice Submission Location
invoice_submission_hour_from	float		yes	Submission From
invoice_submission_hour_to	float		yes	Submission To
client_po_printed_datetime	datetime		yes	Printed Date
client_po_page_count	integer		yes	Page Count
client_po_document_id	many2one	ir_attachment	yes	Scanned PO
total_po_amount_printed	monetary		yes	Total PO Amount (as printed)
tax_table_printed	monetary		yes	Tax Table (as printed)
tax_vat_printed	float		yes	Tax VAT (as printed)
grand_total_printed	monetary		yes	PO Grand Total (as printed)
grand_total_words_printed	char		yes	Amount in Words (as printed)
totals_match	boolean		yes	Totals Match
lines_total_mismatch_count	integer		no	Lines with a Different Total
item_count	integer		yes	Items
items_chosen_count	integer		yes	Items with Vendor
items_obtained_count	integer		yes	Items Obtained
progress_percent	float		yes	Progress (%)
quote_count	integer		no	Quote Count
purchase_count	integer		no	Purchase Count

sale_order_line (columns added)

Field (column)	Type	Links to	Stored	Label in Odoo
client_line_no	integer		yes	S. No.
client_item_code	char		yes	Client Item Code
description_printed	char		yes	Description (as printed)
uom_printed	char		yes	UoM (as printed)
total_printed	monetary		yes	Total (as printed)
line_total_match	boolean		yes	Total Matches
delivery_group_id	many2one	procurement_delivery_group	yes	Delivery Group
quote_ids	one2many	procurement_quote	no	Vendor Quotes
purchase_ids	one2many	procurement_purchase	no	Purchases
chosen_vendor_id	many2one	res_partner	yes	Chosen Vendor
qty_obtained	float		yes	Qty Obtained
qty_remaining	float		yes	Qty Remaining
avg_purchase_price	float		yes	Avg. Purchase Price
margin_amount	monetary		yes	Procurement Margin
procurement_status	selection		yes	Procurement Status

procurement_quote

Field (column)	Type	Links to	Stored	Label in Odoo
sale_line_id	many2one	sale_order_line	yes	Item
order_id	many2one	sale_order	yes	Client PO
client_line_no	integer		yes	S. No.
client_item_code	char		no	Client Item Code
product_id	many2one	product_product	no	Product
item_description	text		no	Description
product_uom_qty	float		no	Qty Ordered
product_uom_id	many2one	uom_uom	no	UoM
vendor_id	many2one	res_partner	yes	Vendor
state	selection		yes	State
quoted_price	monetary		yes	Quoted Unit Price
currency_id	many2one	res_currency	yes	Currency
quote_date	date		yes	Quote Date
notes	char		yes	Notes
price_order_currency	monetary		yes	Price (PO Currency)
order_currency_id	many2one	res_currency	no	PO Currency
is_lowest	boolean		yes	Lowest Price
qty_to_buy	float		yes	Qty to Buy
amount_to_buy	monetary		yes	Total
purchase_ids	one2many	procurement_purchase	no	Purchases
qty_needed	float		no	Qty Needed

procurement_purchase

Field (column)	Type	Links to	Stored	Label in Odoo
sale_line_id	many2one	sale_order_line	yes	Item
order_id	many2one	sale_order	yes	Client PO
client_line_no	integer		yes	S. No.
client_item_code	char		no	Client Item Code
product_uom_id	many2one	uom_uom	no	UoM
qty_ordered	float		no	Qty Ordered
vendor_id	many2one	res_partner	yes	Vendor
purchase_price	monetary		yes	Purchase Unit Price
currency_id	many2one	res_currency	yes	Currency
quantity	float		yes	Quantity
date	date		yes	Date
notes	char		yes	Notes
amount	monetary		yes	Total Cost
state	selection		yes	Status
quote_id	many2one	procurement_quote	yes	Chosen Quote
qty_received	float		yes	Qty Received

procurement_delivery_group

Field (column)	Type	Links to	Stored	Label in Odoo
order_id	many2one	sale_order	yes	Client PO
sequence	integer		yes	Sequence
name	char		yes	Group
description	char		yes	Description
immediate_batch	boolean		yes	Immediate Batch
immediate_batch_weeks	integer		yes	Immediate Batch (weeks)
min_months	integer		yes	Min. Months
max_months	integer		yes	Max. Months
start_event	selection		yes	Start Event
boq_reference	char		yes	BOQ Reference
line_ids	one2many	sale_order_line	no	PO Lines
line_count	integer		no	Lines
immediate_batch_deadline	date		yes	Immediate Batch Deadline
final_deadline	date		yes	Final Deadline

procurement_payment_milestone

Field (column)	Type	Links to	Stored	Label in Odoo
order_id	many2one	sale_order	yes	Client PO
sequence	integer		yes	Sequence
name	char		yes	Milestone
percent	float		yes	Percent (%)

Field (column)	Type	Links to	Stored	Label in Odoo
requires_lg	boolean		yes	Requires LG
lg_type	char		yes	LG Type
trigger	selection		yes	Trigger
collection_days	integer		yes	Collection Days
notes	char		yes	Notes

procurement_quote_request_wizard

Field (column)	Type	Links to	Stored	Label in Odoo
order_id	many2one	sale_order	yes	Client PO
line_ids	one2many	procurement_quote_request_wizard_line	no	Vendors

procurement_quote_request_wizard_line

Field (column)	Type	Links to	Stored	Label in Odoo
wizard_id	many2one	procurement_quote_request_wizard	yes	Wizard
order_id	many2one	sale_order	no	Client PO
vendor_id	many2one	res_partner	yes	Vendor
sale_line_ids	many2many	sale_order_line	link table	Items
all_items	boolean		yes	All Items